reshape:

Change the shape of an array without changing its data

10 20 30 40 50 60

10 20 30
40 50 60

Syntax:

array.reshape(rows, columns)

```
import numpy as np
arr = np.array([10 20 30 40 50 60])
print(new_arr)
```

Output:

[10 20 30
40 50 60]

arr.reshape(4,2) × (not execute) it show error

Automatic reshaping

Instead of Calculating rows numpy can Calculate Automatically using -1

```
arr = np.arrange(1,13)
new_arr = arr.reshape(3,-1)
print(new_arr)
```

Output:

[1 2 3 4
5 6 7 8
9 10 11 12]

Flatten()

[1 2 3
4 5 6]

[1 2 3 4 5 6] → ML Algorithm accept

```
arr=np.array([1 2 3 4 5 6])
print(arr.flatten())
```

Output

[1 2 3 4 5 6]

Copy:

Orginal array not changed

```
np.array([10, 20, 30])
new_arr = arr.copy()
new_arr[0] = 100
Print(arr)
Print(new_arr)
```

Output

[10 20 30]
[100 20 30]

Mathematical operation in array:

Addition

```
arr = np.array([10, 20, 30])
print(arr+5)
```

Output

```
[15 25 35]
```

Sub `(arr-5)` power `(arr**5)` Mul `(arr*.5)`

One array and another array Operation

```
a = np.array([1, 2, 3])
b = np.array([10, 20, 30])
print(a+b)
```

Output:

```
[11 22 33]
```

Broadcasting

Broadcasting Allow numpy array of different shapes (sizes) to perform Arithmetic operations without explicitly making them to same size

```
arr = np.array([10, 20, 30])
print(arr at 5)
```

sym
array + Value
array + array =
array = value
array1 * array2

Building Function

Square root

`print(np.sqrt(arr))` ⇒ square root Universal Building function

`print(np.exp(1, 2, 3))` ⇒ exponential.

`print(np.log(1, 2, 4))` ⇒ logarithm

`print(np.sin(90))` ⇒ trignometry functions

(cos, tan)

Statistical Function

Sum : print (arr. sum ())

Mean : print (arr. Mean ()) ⟹ average

Median : print (arr. Median ()) ⟹ center

Maximum : print (arr. Max ())

Minimum : print (arr. min ())

Standard deviation : print (arr. std ()) ⟹ How spread in data use in Machine le & statistic

matrix multipl } A dot b.

eg.